

ETP formalization benchmark · story vs literal vs two-stage · vacuous laws excluded

Equational Theories Project implications rendered per each run's form, formalized back by each model over OpenRouter, and graded syntactically (checkform.py). Correct = exact, or an accepted symmetry (sides swapped, or both equations uniformly dualized).

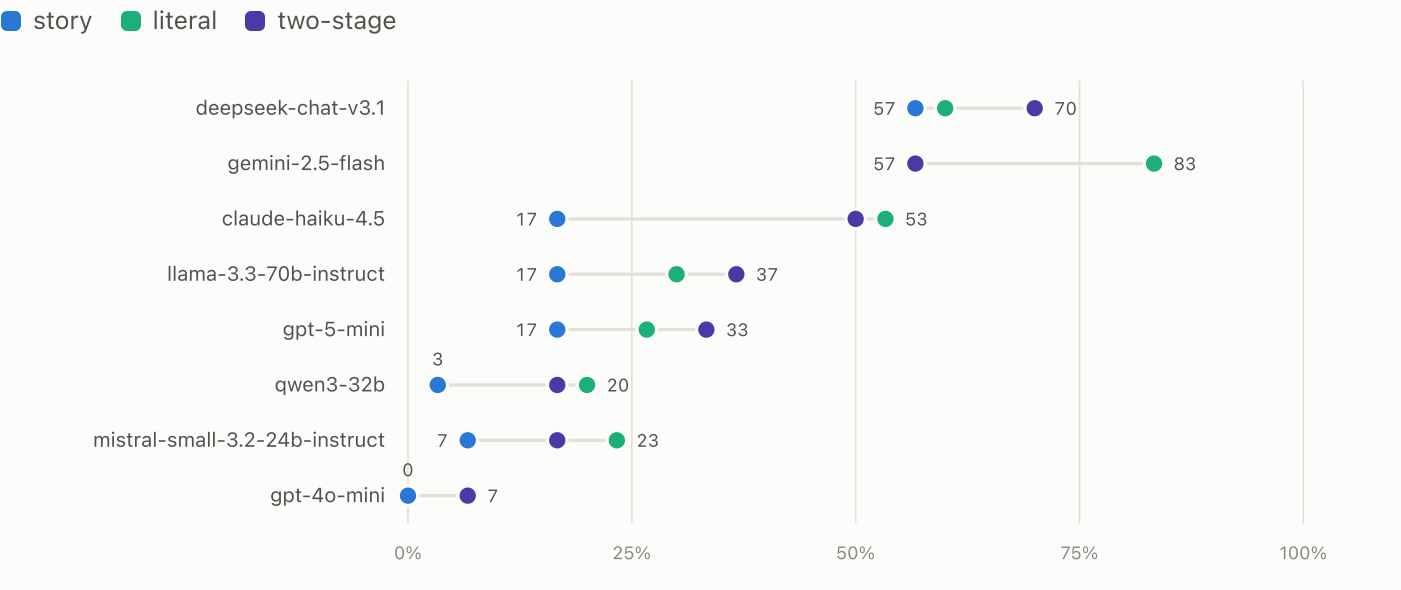
results/no-vacuous/02-reasoning-and-complexity/runs/run-s0-n30-think-off · results/no-vacuous/02-reasoning-and-complexity/runs/run-s0-n30-think-on · results/no-vacuous/02-reasoning-and-complexity/runs/run-strat5-s0-think-off · results/no-vacuous/02-reasoning-and-complexity/runs/run-strat5-s0-think-on · results/no-vacuous/04-structured-literal/runs/run-s0-n30-think-off · results/no-vacuous/04-structured-literal/runs/run-s0-n30-think-on · results/no-vacuous/04-structured-literal/runs/run-strat5-s0-think-off · results/no-vacuous/04-structured-literal/runs/run-strat5-s0-think-on · results/no-vacuous/05-two-stage/runs/run-s0-n30-think-off · results/no-vacuous/05-two-stage/runs/run-s0-n30-think-on · results/no-vacuous/05-two-stage/runs/run-strat5-s0-think-off · results/no-vacuous/05-two-stage/runs/run-strat5-s0-think-on

Runs	Models	Stories	Graded calls	Overall correct
12	8	54	2,592	69%

30 uniformly sampled pairs · seed 0 · story vs literal vs two-stage · no-think and all-think

FIGURE 1 Correct rate by model and form — no-think · uniform 30

The same pairs rendered per arm; each row compares one model across the forms under the no-think regime. Sample: 30 uniformly drawn pairs — almost entirely maximal 4+4-operation implications.

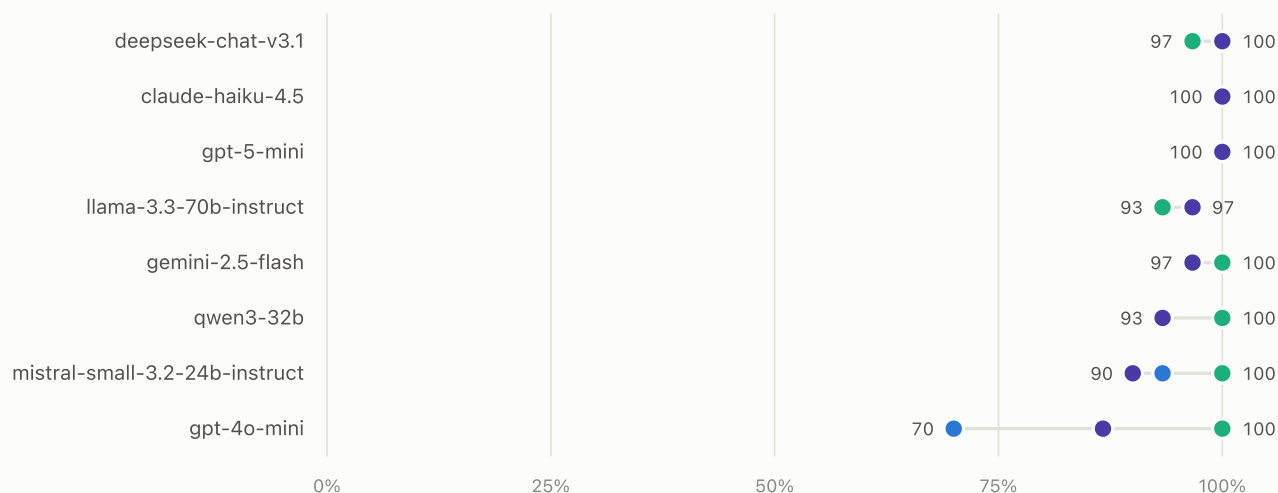


► Data: correct rate per model and form

FIGURE 2 **Correct rate by model and form — all-think · uniform 30**

The same pairs rendered per arm; each row compares one model across the forms under the all-think regime. Sample: 30 uniformly drawn pairs — almost entirely maximal 4+4-operation implications.

■ story ■ literal ■ two-stage

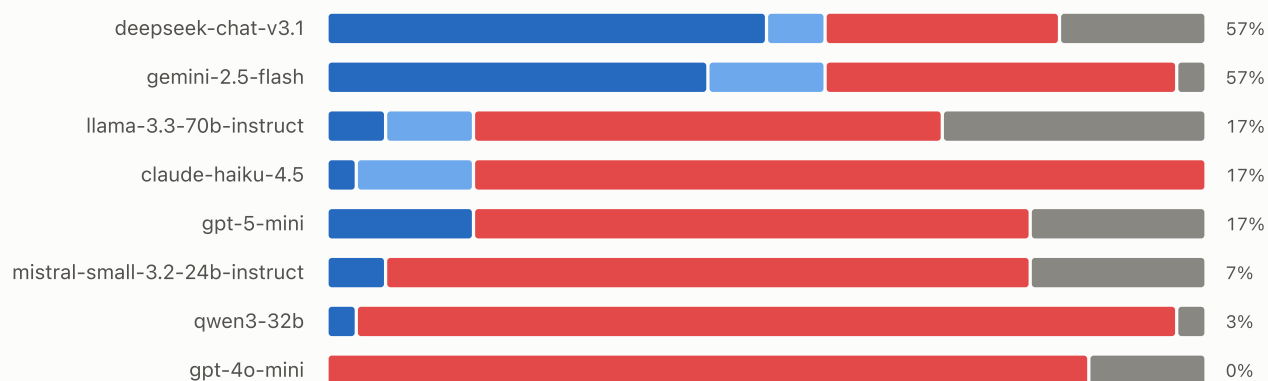


► Data: correct rate per model and form

FIGURE 3 **Verdict composition — story · no-think · uniform 30**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 30 uniformly drawn pairs — almost entirely maximal 4+4-operation implications.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable

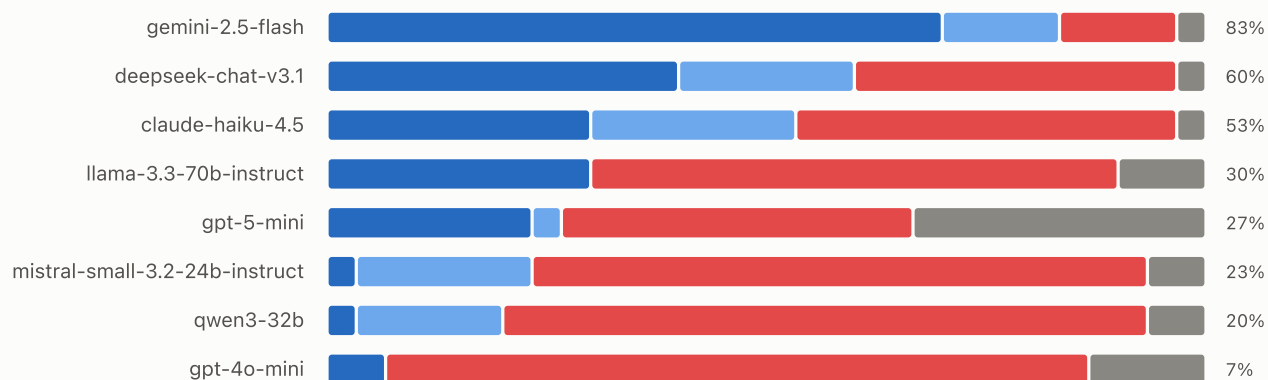


► Data: verdict counts per model

FIGURE 4 **Verdict composition — literal · no-think · uniform 30**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 30 uniformly drawn pairs — almost entirely maximal 4+4-operation implications.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable

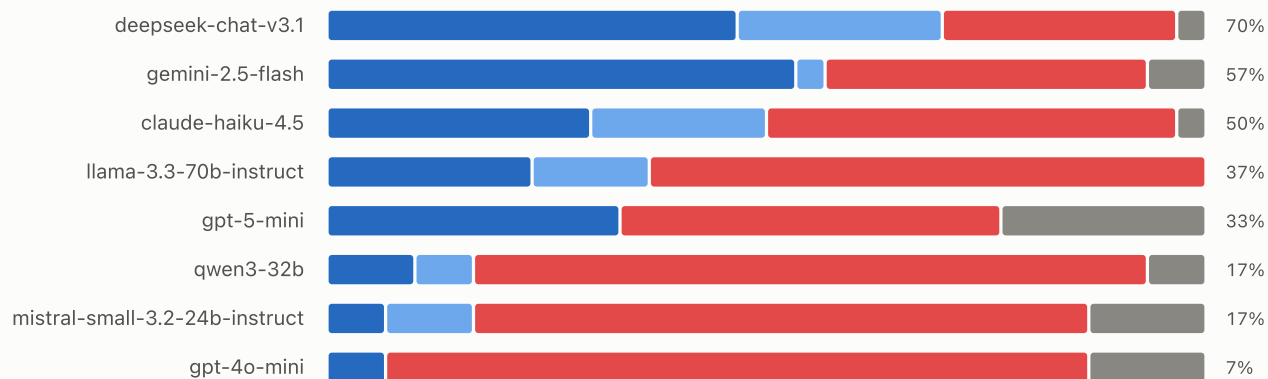


► Data: verdict counts per model

FIGURE 5 **Verdict composition — two-stage · no-think · uniform 30**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 30 uniformly drawn pairs — almost entirely maximal 4+4-operation implications.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model

FIGURE 6 **Verdict composition — story · all-think · uniform 30**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 30 uniformly drawn pairs — almost entirely maximal 4+4-operation implications.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model

FIGURE 7 **Verdict composition — literal · all-think · uniform 30**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 30 uniformly drawn pairs — almost entirely maximal 4+4-operation implications.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable

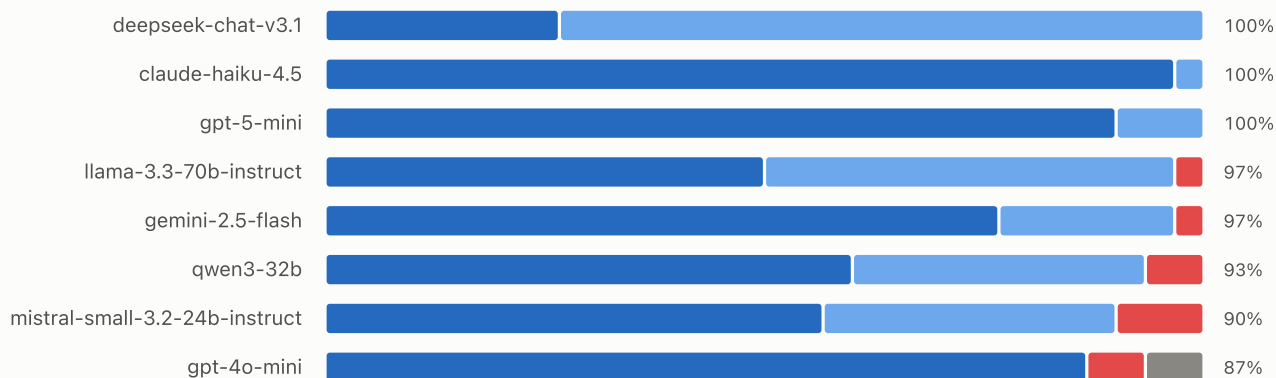


► Data: verdict counts per model

FIGURE 8 **Verdict composition — two-stage · all-think · uniform 30**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 30 uniformly drawn pairs — almost entirely maximal 4+4-operation implications.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



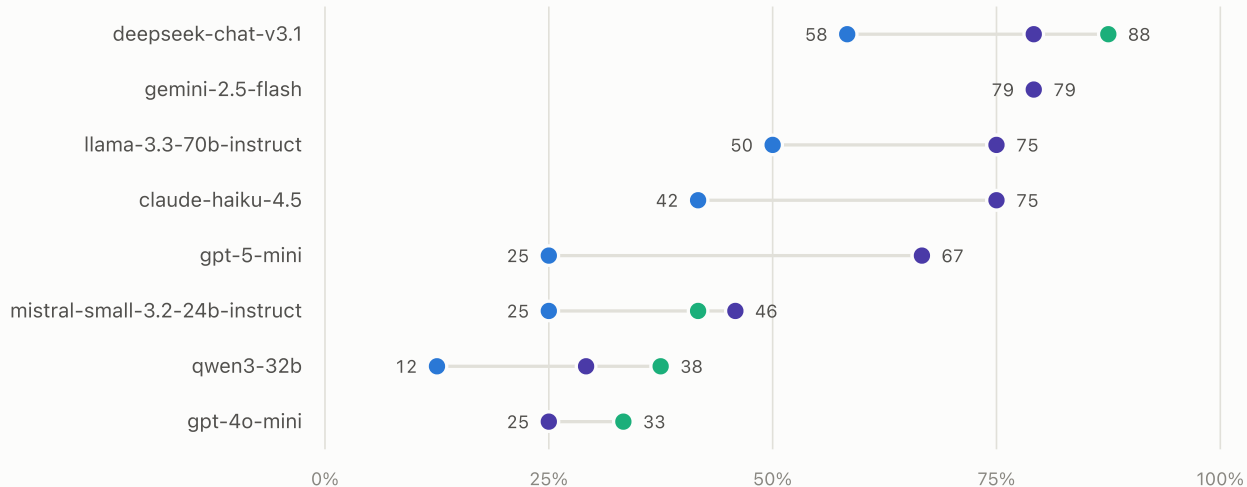
► Data: verdict counts per model

24 pairs, 5 per operation-count bin 1–8 · seed 0 · story vs literal vs two-stage · no-think and all-think

FIGURE 9 **Correct rate by model and form — no-think · stratified 24**

The same pairs rendered per arm; each row compares one model across the forms under the no-think regime.
Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.

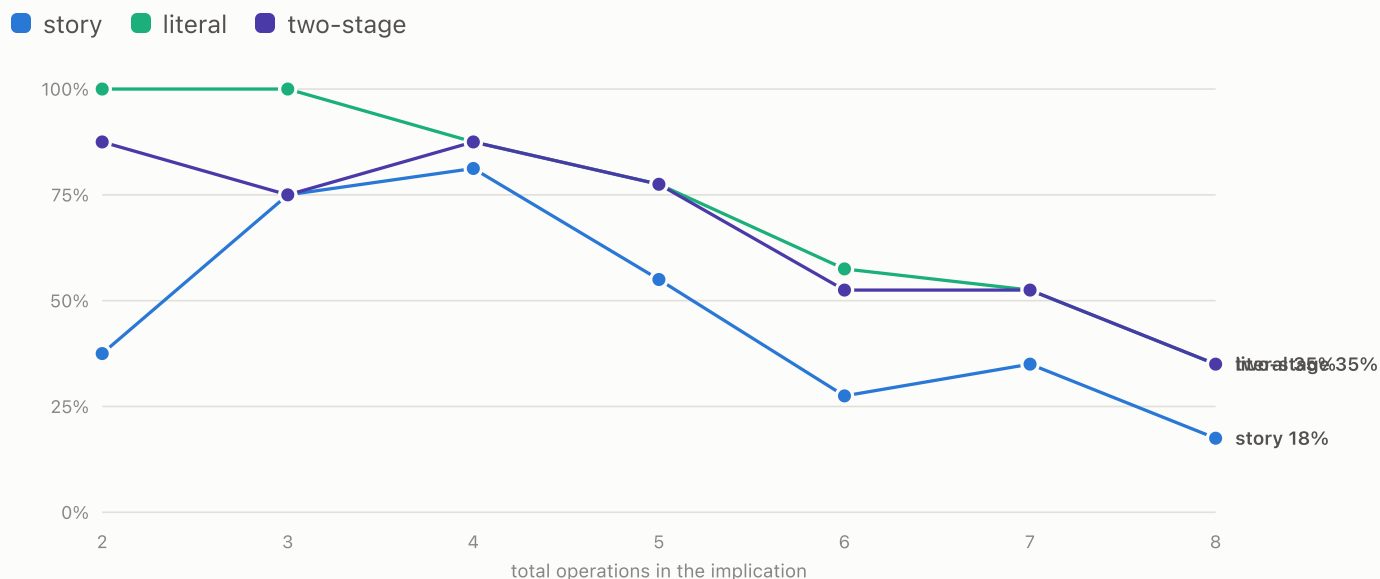
■ story ■ literal ■ two-stage



► Data: correct rate per model and form

FIGURE 10 **Accuracy vs operation count — no-think · stratified 24**

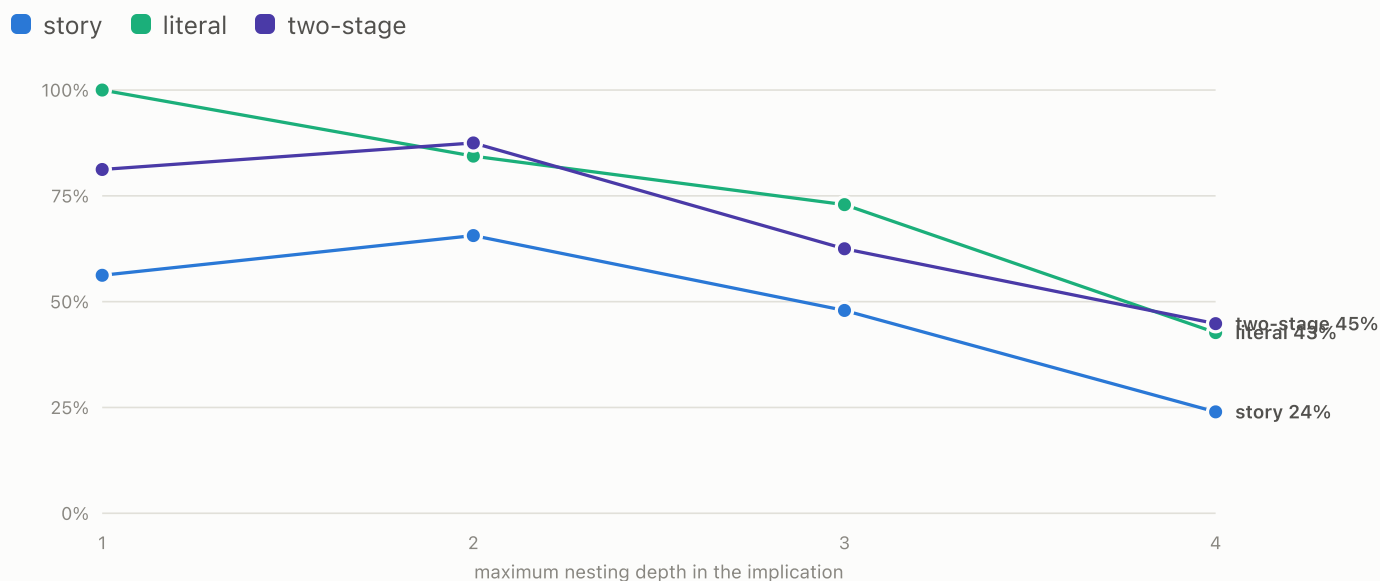
Correct rate pooled over all models, by the implication's total operation count (both equations combined); one line per form, no-think regime. Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.



► Data: pooled correct rate per operation-count bin

FIGURE 11 **Accuracy vs nesting depth — no-think · stratified 24**

Correct rate pooled over all models, by the deepest operation nesting across the four equation sides; one line per form, no-think regime. Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.

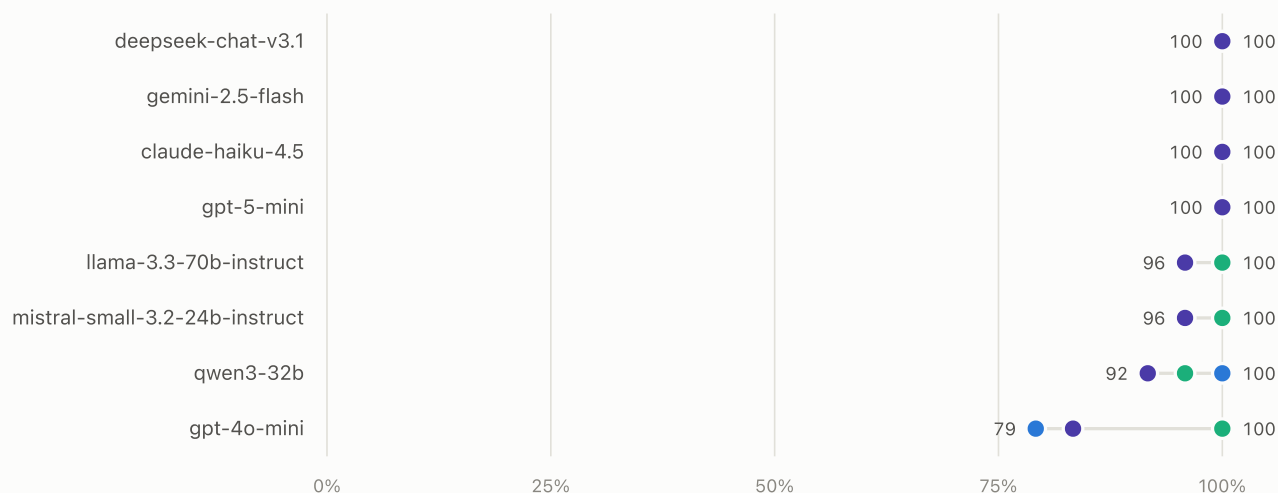


► Data: pooled correct rate per depth bin

FIGURE 12 **Correct rate by model and form — all-think · stratified 24**

The same pairs rendered per arm; each row compares one model across the forms under the all-think regime. Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.

■ story ■ literal ■ two-stage

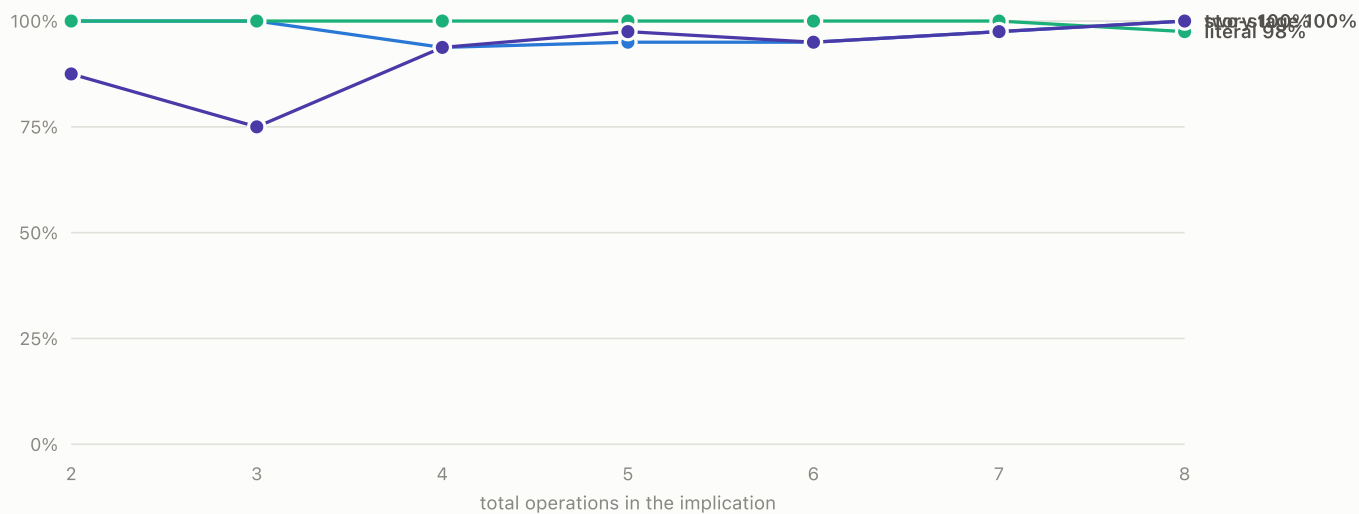


► Data: correct rate per model and form

FIGURE 13 **Accuracy vs operation count — all-think · stratified 24**

Correct rate pooled over all models, by the implication's total operation count (both equations combined); one line per form, all-think regime. Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.

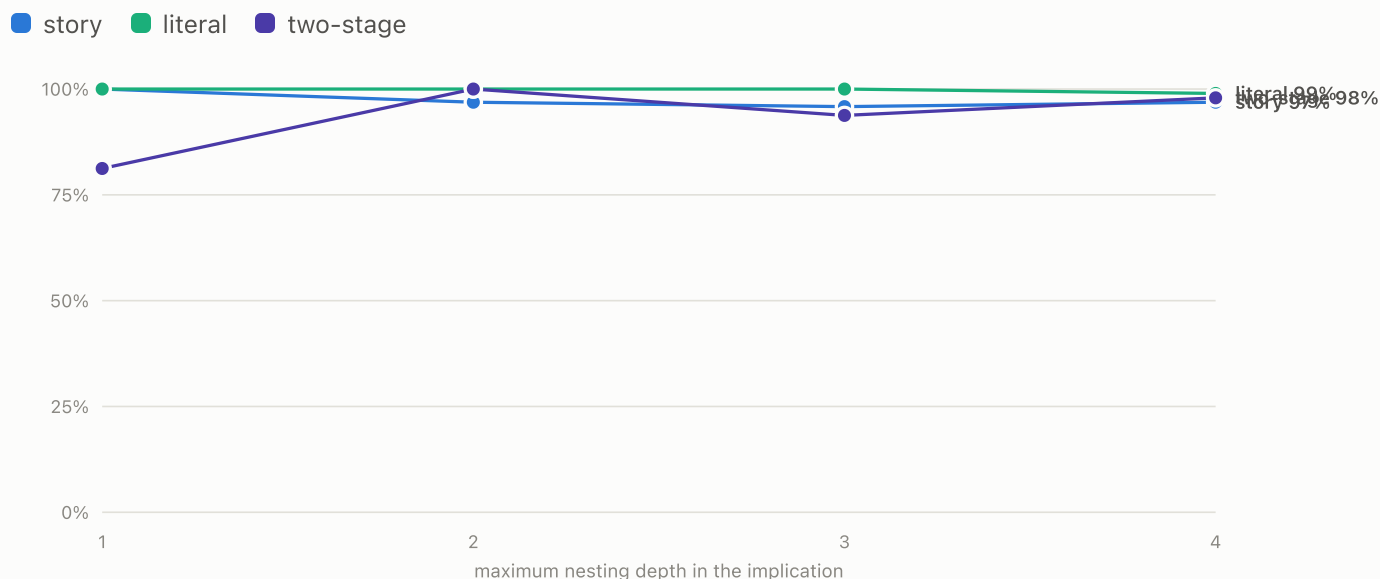
■ story ■ literal ■ two-stage



► Data: pooled correct rate per operation-count bin

FIGURE 14 **Accuracy vs nesting depth — all-think · stratified 24**

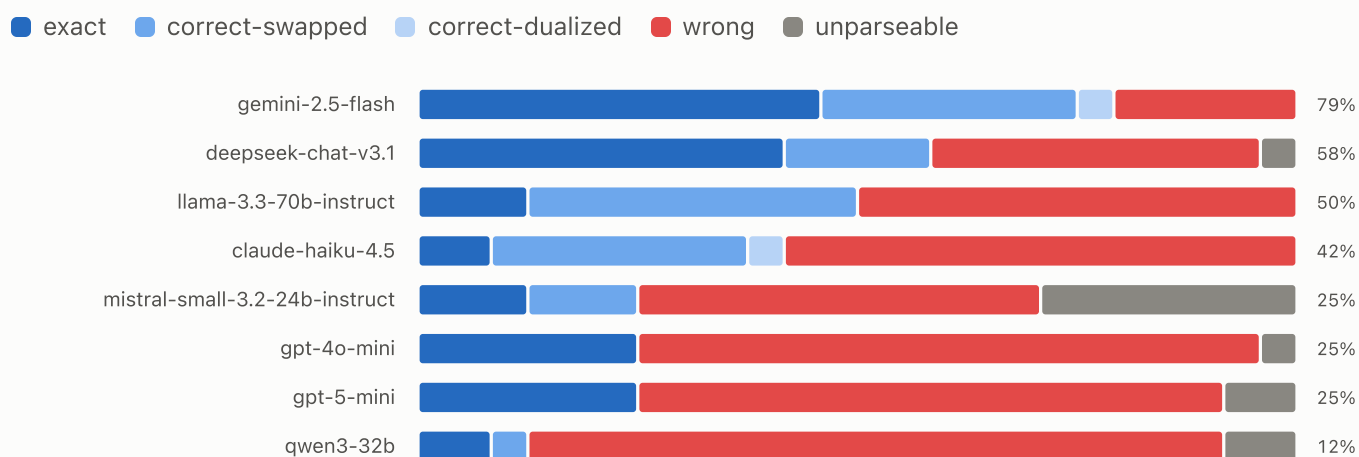
Correct rate pooled over all models, by the deepest operation nesting across the four equation sides; one line per form, all-think regime. Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.



► Data: pooled correct rate per depth bin

FIGURE 15 **Verdict composition — story · no-think · stratified 24**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.

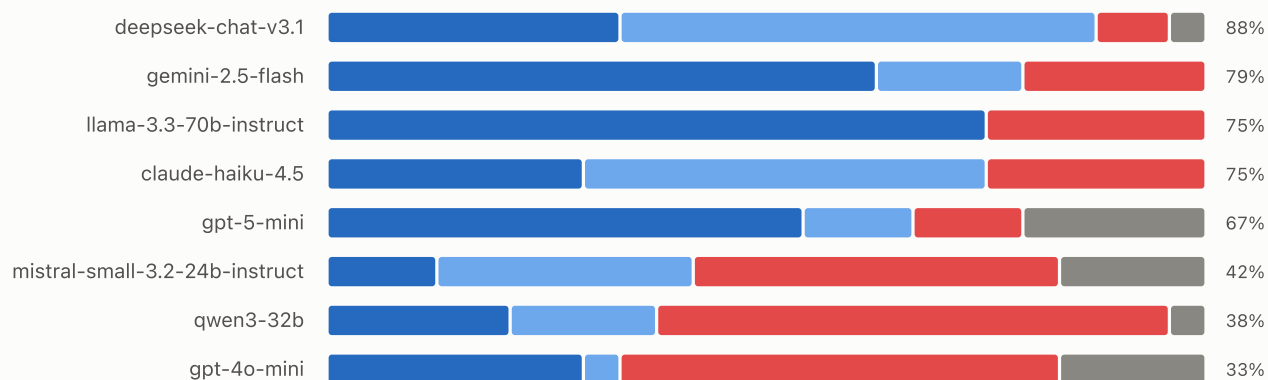


► Data: verdict counts per model

FIGURE 16 **Verdict composition — literal · no-think · stratified 24**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable

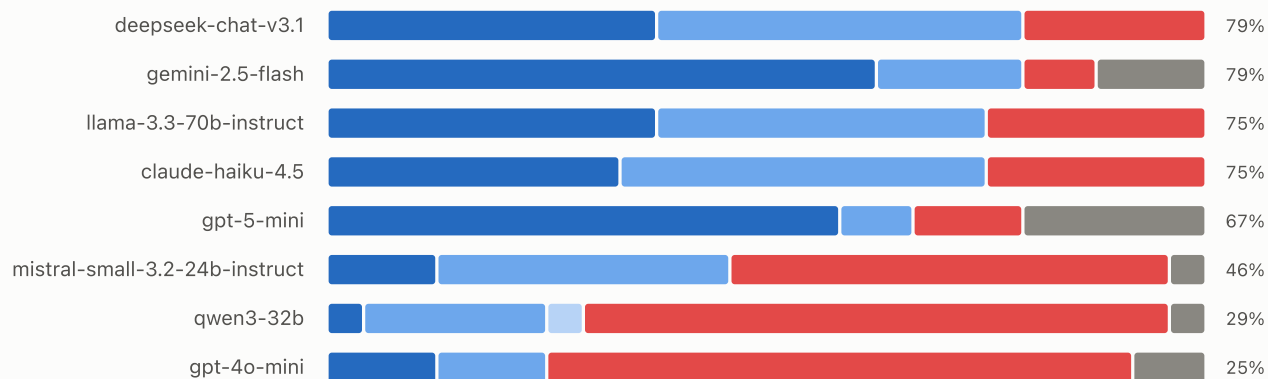


► Data: verdict counts per model

FIGURE 17 **Verdict composition — two-stage · no-think · stratified 24**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.

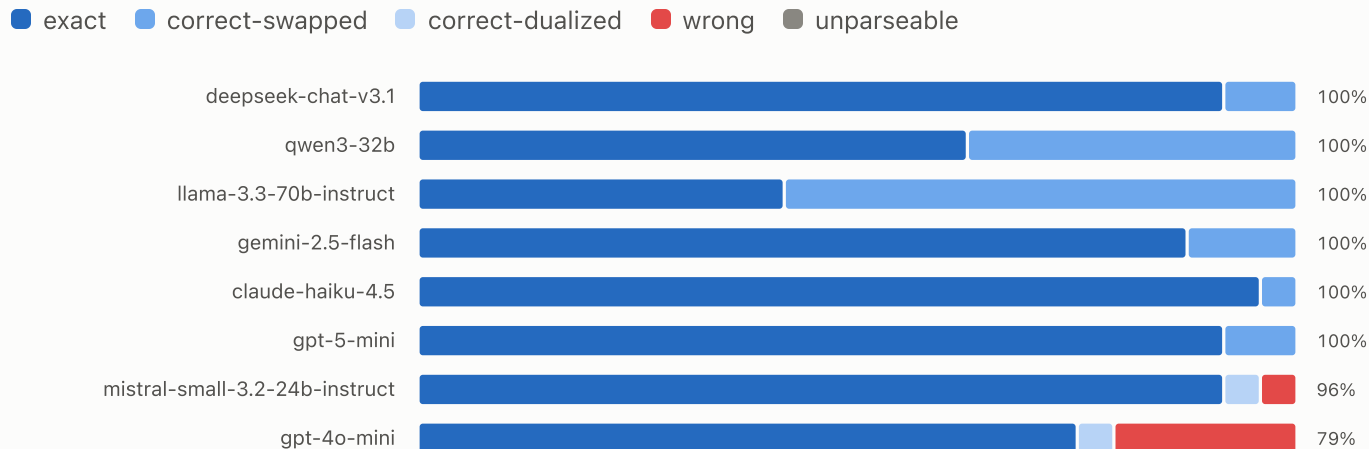
■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model

FIGURE 18 **Verdict composition — story · all-think · stratified 24**

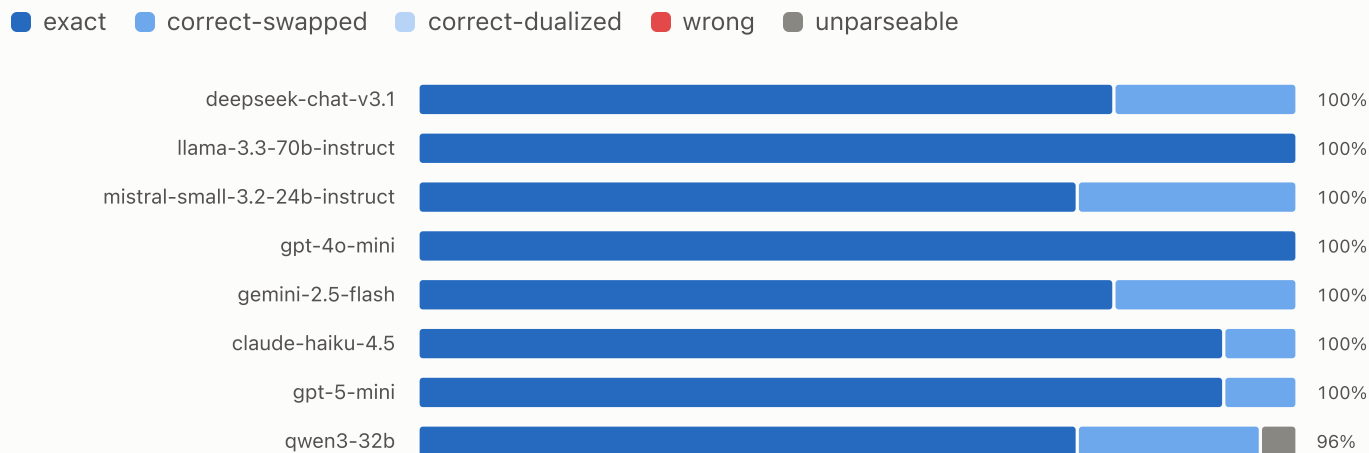
How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.



► Data: verdict counts per model

FIGURE 19 **Verdict composition — literal · all-think · stratified 24**

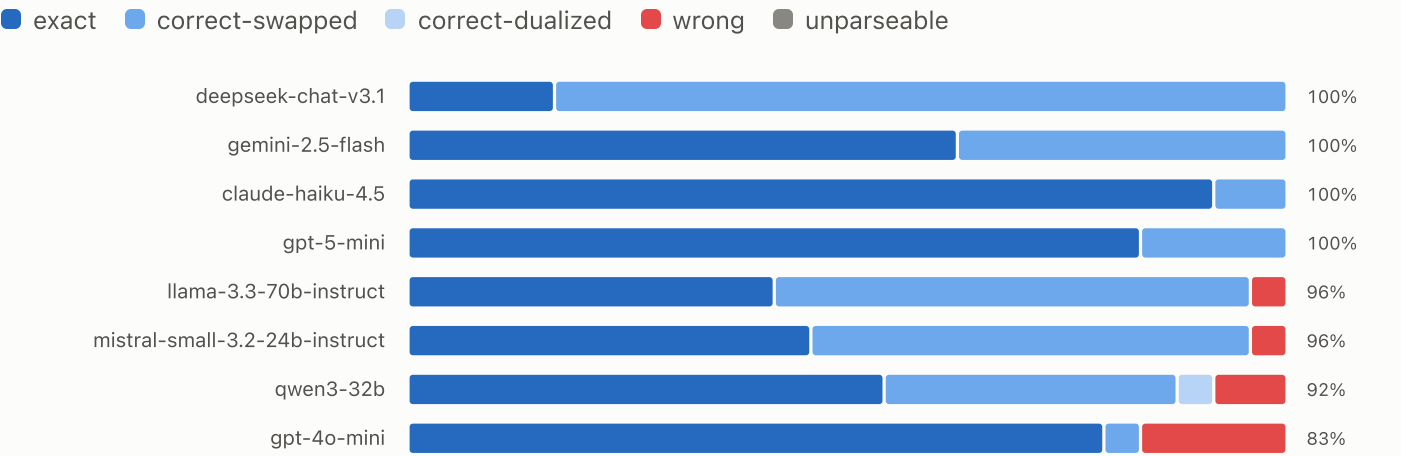
How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.



► Data: verdict counts per model

FIGURE 20 **Verdict composition — two-stage · all-think · stratified 24**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 24 pairs, 5 per total-operation bin 1–8 — the full complexity range, including trivial laws.



► Data: verdict counts per model